For example, $A = \begin{pmatrix} \sin x \\ \cos x \\ 1 \end{pmatrix}$, $B = \begin{pmatrix} \sqrt{5} \\ 7 \end{pmatrix}$ and $C = \begin{pmatrix} 8 \\ -3 \\ 23 \\ 17 \end{pmatrix}$ are column matrices of order 3×1, 2×1 and 4×1 respectively.

In general, $A = \begin{pmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ a_{m1} \end{pmatrix}$ is a column matrix of order $m \times 1$.

3. Square Matrix

A matrix in which the number of rows is equal to the number of columns is called a square matrix. Thus a matrix $A = (a_{ij})_{m \times n}$ will be a square matrix if $m = n$

For example, $\begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}_{2 \times 2}$, $\begin{pmatrix} -1 & 0 & 2 \\ 3 & 6 & 8 \\ 2 & 3 & 5 \end{pmatrix}_{3 \times 3}$ are square matrices.

In general, $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}_{2 \times 2}$, $\begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix}$ are square matrices of orders 2×2 and 3×3 respectively.

$A = (a_{ij})_{m \times m}$ is a square matrix of order m.

Definition : In a square matrix, the elements of the form a_{11}, a_{22}, a_{33}, . . . (i.e) a_{ii} are called leading diagonal elements. For example in the matrix $\begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}$, 1 and 5 are leading diagonal elements.

4. Diagonal Matrix

A square matrix, all of whose elements, except those in the leading diagonal are zero is called a diagonal matrix.

(ie) A square matrix $A = (a_{ij})$ is said to be diagonal matrix if $a_{ij} = 0$ for $i \neq j$. Note that some elements of the leading diagonal may be zero but not all.

For example, $\begin{pmatrix} 8 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 11 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ are diagonal matrices.

5. Scalar Matrix

A diagonal matrix in which all the leading diagonal elements are equal is called a scalar matrix.

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